NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge Factsheet Perceptual Score-Guided Image Super-resolutuon for Mobile Real-World Applications

Tongqi Guo, Ruhan Liu, Sangzi Shi, Huazhang Deng, Jie Yang
Wenzhuo Ma, Yuantong Zhang, Daiqin Yang, Zhenzhong Chen*
Wuhan University
{2024302131019,ruhan.liu,2024302131080,2023302131136,yang-jie}@whu.edu.cn
{mwzgorgeous,yuantongzhang,dqyang,zzchen}@whu.edu.cn
*Corresponding author

1. Factsheet Information

1.1. Team details

- Team name: Antman
- Team leader name: Zhenzhong Chen
- Team leader address, phone number, and email: School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, 430079, China, +86 18162614523, zzchen@whu.edu.cn
- Rest of the team members: Tongqi Guo, Ruhan Liu, Sangzi Shi, Huazhang Deng, Jie Yang, Wenzhuo Ma, Yuantong Zhang, Daiqin Yang
- Affiliation: Wuhan University, China
- Affiliation of the team and/or team members with NTIRE 2026 sponsors (check the workshop website): None
- User names and entries on the NTIRE 2026 CodaBench competitions (development/validation and testing phases): ant-man, https://www.codabench.org/profiles/user/ant-man/
- Best scoring entries of the team during the testing phase: https://www.codabench.org/competitions/13509/detailed_results/628600/
- Link to the codes/executables of the solution(s): https://github.com/gunshi3/NTIRE2026_Mobile_RealWorld_ImageSR
- Link to the final results of the solution(s) on the testing data: https://drive.google.com/file/d/1MvjM8NUqXavxYdJzVJTlnridStCoa1ct/view?usp=drive_link

1.2. Method details

Method. Our solution is built on the RRDBNet(Fig.1) generator in ESRGAN[5], and we initialize the model with the pretrained Real-ESRGAN_x4plus weights[6]. The network keeps the standard Real-ESRGAN_x4plus configuration, including 3 input and output channels, 64 feature channels, 32 growth channels, and 23 RRDB blocks. This design provides sufficient representation capacity for restoring real-world degradations while maintaining a practical inference cost for the challenge setting.

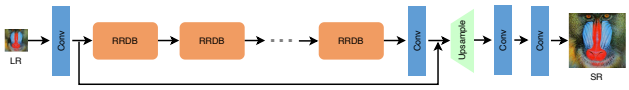

Figure 1. The pipeline of RRDBNet.

Optimization. To adapt the pretrained generator to the target track, we combine a pixel reconstruction objective with perceptual score guidance. The reconstruction term is the L_1 loss between the restored image and the ground truth. In addition, we introduce a weighted score-guided loss composed of three no-reference quality predictors, namely MANIQA[7], MUSIQ[3], and CLIPIQA[4], together with two full-reference perceptual distance metrics, LPIPS[8] and DISTS[1]. During optimization, the model is encouraged to produce higher MANIQA, MUSIQ, and CLIPIQA scores while reducing LPIPS and DISTS. All quality scores are computed on clamped outputs in the range $[0, 1]$ for stable training.

Training Details. We fine-tune the model on DIV2K[2]. Specifically, we use the 800-image unknown-degradation training split of DIV2K. HR patches of size 192×192 are randomly cropped, and random flipping is used for data augmentation. The mini-batch size is set to 4. We optimize the network with Adam for 10,000 iterations, starting from a learning rate of 5×10^{-6} and halving it at 9,000 iterations. An exponential moving average (EMA) strategy is used throughout training.

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